ENGINEERING MATHEMATICS - II

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UNIT 5 : Fourier series
Session : 5
Sub Topic : Half Range Fourier Series

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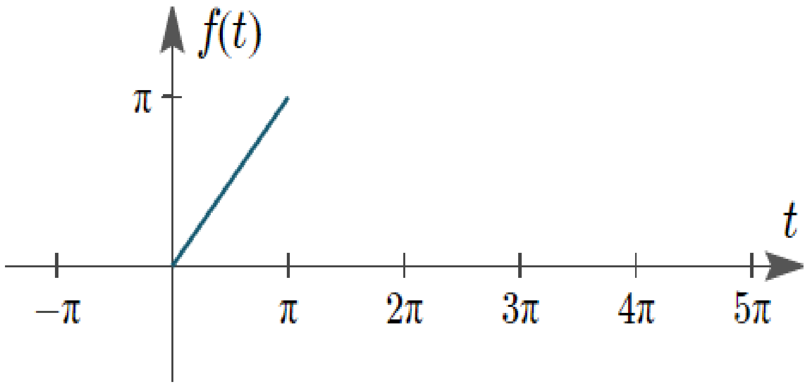
Half Range Fourier series

- Consider a function $f(x)$ for which the Fourier series is required is defined over half the range from 0 to π (or $0\ to\ l$).
- In such cases, the given function can be extended to the other half range, as either an odd function or an even function.
- The Fourier expansion of such a function will consist either only sine or only cosine terms.

The series produced is then called a **half range Fourier series**.

Example:

Consider the function $f(t) = t, 0 < t < \pi$ to be half of a period π of a 2π periodic function. The extension can be done to the other half either as an odd function or as an even function.

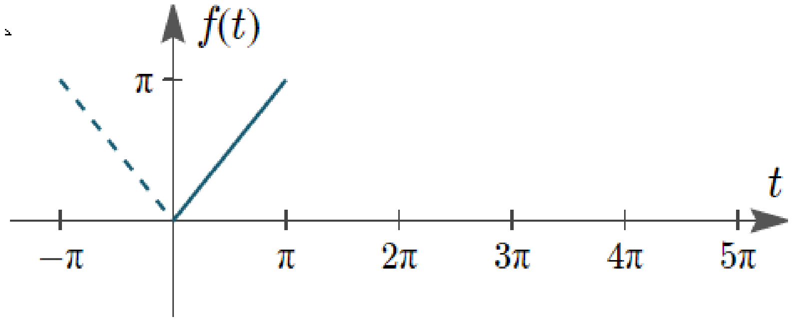


Graph of $f(t)$.

Half Range Fourier series

Case (i) Even Extension

We will extend this function to the other half of the range $(-\pi, 0)$ as an even function.



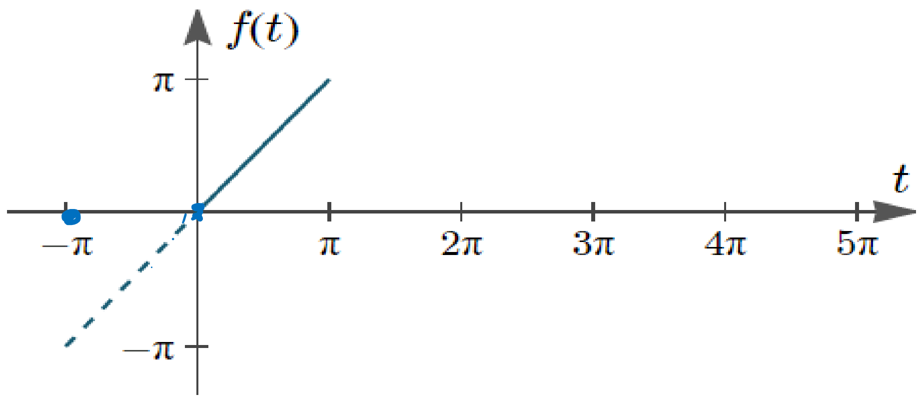
Graph of $f(t)$, illustrating it's an even function.

So we have an even function in the interval $(-\pi, \pi)$ and the fourier series expansion will consists of only cosine terms and is called Half Range Fourier Cosine Series.

Half Range Fourier series

Case (ii) Odd Extension

We will extend this function to the other half of the range $(-\pi, 0)$ as an odd function.



Graph of $f(t)$, showing it's an odd function.

So we have an odd function in the interval $(-\pi, \pi)$ and the Fourier series expansion will consists of only sine terms and is called Half Range Fourier Sine Series.

Half Range Fourier series of $f(x)$ in $(0, \pi)$

Half Range Cosine series	Half Range Sine Series
$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$	$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$
Fourier Coefficients:	**Fourier Coefficients:**
❖ $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx.$	❖ $b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \sin nx \, dx$
❖ $a_n = \frac{2}{\pi}\int_0^{\pi} f(x) \cos nx \, dx$	

Half Range Fourier series of $f(x)$ in $(0, l)$

Half Range Cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l}$$

Fourier Coefficients:

- $a_0 = \frac{2}{l}\int_0^l f(x)dx.$
- $a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$

Half Range Sine Series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$$

Fourier Coefficients:

- $b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$

Problems on Half Range Fourier series

1. Find the Fourier half range a) Cosine series b) Sine series of

$$\mathbf{f}(\boldsymbol{x}) = \begin{cases} \boldsymbol{x}, & \mathbf{0} < \boldsymbol{x} < \mathbf{1} \\ \mathbf{2} - \boldsymbol{x}, & 1 < \boldsymbol{x} < \mathbf{2} \end{cases}$$

Solution:

a) Cosine series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n cos \frac{n\pi x}{l}$

Comparing (0,2) with half the range $(0,l)$, we have $l = 2$

Therefore $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n cos \frac{n\pi x}{2}$

$$\boldsymbol{a_0} = \frac{\mathbf{2}}{\mathbf{2}} \int_{\mathbf{0}}^{\mathbf{2}} \boldsymbol{f(x)dx}$$

$$= \int_0^1 x\, dx + \int_1^2 (2 - x)\mathrm{dx}$$

Problems on Half Range Fourier series

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$$a_0 = \left[\frac{x^2}{2}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^2$$

$$\boldsymbol{a_0 = 1}$$

$$\boldsymbol{a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx}$$

$$a_n = \frac{2}{2}\int_0^2 f(x)\cos\frac{n\pi x}{2}\,dx$$

$$= \int_0^1 x\cos\frac{n\pi x}{2}\,dx + \int_1^2 (2-x)\cos\frac{n\pi x}{2}\,dx$$

$$= \left[x.\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_0^1 - \left[-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_0^1 + \left[(2-x).\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_1^2 - \left[\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_1^2$$

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$$= \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)+\frac{4}{n^2\pi^2}\left[cos\left(\frac{n\pi}{2}\right)-1\right]-\frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)-\frac{4}{n^2\pi^2}\left[cosn\pi-cos\left(\frac{n\pi}{2}\right)\right]$$

$$a_n=\frac{8}{n^2\pi^2}cos\left(\frac{n\pi}{2}\right)-\frac{4}{n^2\pi^2}[1+(-1)^n]$$

$\therefore$ **Cosine series is given by**

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n cos\frac{n\pi x}{2}$$

$$f(x)=\frac{1}{2}+\sum_{n=1}^{\infty}\frac{8}{n^2\pi^2}\ cos\left(\frac{n\pi}{2}\right)-\frac{4}{n^2\pi^2}[1+(-1)^n]cos\frac{n\pi x}{2}$$

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b) Sine series : $f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$

where $b_n = \frac{2}{l}\int_0^l f(x)\, sin\frac{n\pi x}{l}\, dx$

Comparing (0,2) with half the range $(0,l)$, we have $l = 2$

$$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{2}$$

$$b_n = \frac{2}{2}\int_0^2 f(x) sin\frac{n\pi x}{2} dx$$

$$= \int_0^1 x\, sin\frac{n\pi x}{2} dx + \int_1^2 (2-x) sin\frac{n\pi x}{2} dx$$

$$= \left[x.\frac{-\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_0^1 - \left[-\frac{s\,in\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_0^1 + \left[(2-x).\frac{-cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_1^2 - \left[\frac{\sin\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_1^2$$

Problems on Half Range Fourier series

$$b_n = \frac{-2}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2}\left[sin\left(\frac{n\pi}{2}\right)\right] + \frac{2}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2}\left[0 - sin\left(\frac{n\pi}{2}\right)\right]$$

$$b_n = \frac{8}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)$$

$\therefore$ Sine series is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sin\frac{n\pi x}{2}$$

THANK YOU

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